

Skills Summary

Rules, Tables, Graphs, and Sensible Domains

Use this reference while completing your worksheet or when studying.

Skill 1 — From a rule to a table of values

Rule: the input goes in, the output comes out. To fill a table, substitute each x into the rule — in parentheses, so signs and squares survive — and record the result under it.

Example: Build a table for $f(x) = 4x - 1$ at $x = -1$ and $x = 2$.

Step 1. Each table entry is an evaluation: the top row holds inputs, the bottom row holds the outputs the rule produces.

Step 2. $4(-1) - 1 = -4 - 1$ and $4(2) - 1 = 8 - 1$.

$$f(-1) = -5 \quad \text{and} \quad f(2) = 7$$

Try it: For $g(x) = 5x - 2$, find $g(3)$.

check: $g(3) = 13$

Skill 2 — Solving when the output is given

Two different questions. “Find $f(5)$ ” hands you the *input*: substitute. “Solve $f(x) = 5$ ” hands you the *output*: set the rule equal to 5 and undo it one operation at a time.

On a table, the first reads across from a value in the top row; the second hunts backwards from a value in the bottom row.

Example: For $f(x) = 7x + 4$, solve $f(x) = 39$.

Step 1. The output is given and the input is unknown, so this is an equation to solve, not a substitution.

Step 2. $7x + 4 = 39$; subtract 4 to get $7x = 35$, then divide by 7.

$$x = 5$$

Try it: For $f(x) = 9 - 2x$, solve $f(x) = 1$.

check: $f(4) = 1$

Skill 3 — From a rule to a graph

Rule: every row of the table is one point $(x, f(x))$ on the graph. Plot two or three of them and join them — for a linear rule the line through them is the whole graph.

Reading it back: the height where the graph meets the vertical axis is the output at input 0; the place it crosses the horizontal axis is the input whose output is 0.

Example: Graph $f(x) = 2x - 6$ and find where it crosses the horizontal axis.

Step 1. Two points fix a line, so evaluate at a convenient input first, then find the crossing by solving for the input that gives output zero.

Step 2. $f(0) = 2(0) - 6$, which places the point $(0, -6)$ on the graph.

Step 3. Crossing: $2x - 6 = 0$, so $2x = 6$.

$f(0) = -6$ and the graph crosses at $x = 3$

Try it: For $f(x) = 3x - 12$, find $f(0)$ and where the graph crosses the horizontal axis.

check: $f(0) = -12$, crossing at $x = 4$

Skill 4 — Which inputs the situation allows

Rule: an algebraic rule accepts every number, but a model does not. Ask what the input measures: counts must be whole numbers, lengths and times cannot be negative, and a physical limit (seats, capacity, the moment a tank empties) caps the top end.

Example: A pool holds 300 litres and drains 25 litres a minute, so $W(t) = 300 - 25t$ litres after t minutes. Find $W(4)$, and the moment the pool is empty.

Step 1. Time cannot run backwards and the pool cannot hold less than nothing, so the model only lives from the start until the pool empties — find both ends.

Step 2. $W(4) = 300 - 25(4) = 300 - 100$.

Step 3. Empty means output zero: $300 - 25t = 0$, so $25t = 300$. The sensible inputs are the times from 0 to that moment.

$W(4) = 200$ litres, empty at $t = 12$ minutes

Try it: $V(t) = 150 - 15t$ litres after t minutes. Find $V(4)$ and the time the tank is empty.

check: $V(4) = 90$, empty at $t = 10$